

A Review of Voltage Control Studies on Low Voltage Distribution Networks Containing High Penetration Distributed Photovoltaics

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Abstract: Distributed photovoltaic (PV) in the distribution network accounted for an increasing proportion of the distribution network, and the power quality of the distribution network of the power quality problem is more and more significant. In this paper, the voltage regulation methods for low-voltage distribution networks containing high-penetration PV are investigated. First, the working principles of the four voltage control methods are introduced: energy storage system configuration, regulating the reactive power output of PV inverters, restricting the active power output of PV, adjusting the switching positions of on-load regulator trap changer and distribution network reconfiguration, and then, in combination with the recent related research, the optimization of each method is compared horizontally with its respective concerns and characteristics. The optimization of each method is compared horizontally with the recent studies to find out the focus and characteristics of each method, and the shortcomings of each method are explored. Coordinated voltage control through multiple flexibility resources has become the mainstream voltage regulation scheme, and distribution network voltage regulation is considered from the perspective of flexibility resources. The three types of flexibility resources, namely, source, network, and storage, have been widely used in distribution network voltage regulation. Although load-side resources have become one of the main regulation resources of the new type of power system, the current study introduces less about the participation of load-side flexibility resources in voltage regulation of LV distribution networks and advancing the application of load-side resources in voltage regulation of LV distribution networks is the focus of future research. Then, the important role of load-side flexibility resources in voltage regulation is described in three parts, namely, the important role of load-side resources, the development trend, and the suggestions for promoting the coordination of source-network-load-storage flexibility resources, aiming to promote the application of load-side resources in voltage regulation in LV distribution networks, and the suggestions and programs are proposed for the technological challenges faced by voltage regulation. In the context of today's new power system emphasizing the interaction of source, network, load, and storage, new technologies and methods for solving voltage problems in LV distribution networks are prospected, with a view to providing certain reference value for the actual operation and optimization of distribution network systems.

Keywords: distributed photovoltaic; voltage overrun; distribution network; voltage regulation



Citation: Gao, X.; Zhang, J.; Sun, H.; Liang, Y.; Wei, L.; Yan, C.; Xie, Y. A Review of Voltage Control Studies on Low Voltage Distribution Networks Containing High Penetration Distributed Photovoltaics. *Energies* **2024**, *17*, 3058. <https://doi.org/10.3390/en17133058>

Academic Editor: Georgios Christoforidis

Received: 10 May 2024

Revised: 28 May 2024

Accepted: 19 June 2024

Published: 21 June 2024



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1. Introduction

With the gradual increase in global demand for clean energy, photovoltaic (PV) power generation, as a representative of renewable energy, is ushering in a moment of rapid development. China has firmly established itself as the leader of the global PV market, and according to data from the National Energy Administration (NEA), a new grid-connected capacity of 216.30 GW will be added in 2023, a year-on-year increase of 148%, equivalent to the sum of the new domestic installed capacity from 2019 to 2022. Among them, 120.014 GW of centralized PV power plants and 96.286 GW of distributed PV, while the installed capacity

of household PV among distributed PV reached 43.483 GW. The cumulative grid-connected capacity as of the end of 2023 was 608.918 GW, of which 354.481 GW was centralized PV power plants, 254.438 GW was distributed PV power plants, and 115.797 GW was household PV.

Distributed PV has the characteristics of small investment scale, fast construction speed, low entry threshold, wide development scope, and many subjects involved, etc. Driven by policy support, technological progress, etc., it will still maintain rapid growth in the future [1,2]. It is expected that by 2030, the installed capacity of distributed PV in China will exceed 500 GW.

At the international level, the PV markets in Europe, the United States, and Asia are also showing a booming trend, with the global PV new installations at 444 GW in 2023, up 76% year-on-year, with half of the incremental growth coming from China. The total global PV installations reached 1552.3 GW in 2023 and are predicted to reach 1954.6 GW in 2024, with 402.3 GW of new installations. The European Green New Deal is an important driver for the development of PV and other clean technologies. Many European countries are committed to increasing the proportion of renewable energy in the overall energy mix and promoting the widespread use of green energy.

However, with the increased penetration of PV systems, power quality problems have become significant, especially in distributed PV systems, where the imbalance between PV output and load leads to reverse currents in the lines, causing voltage overruns. For the voltage overrun caused by distributed PV grid-connected, voltage regulation becomes a key factor affecting the stability of the system. In this paper, we will deeply analyze the current status of PV development at home and abroad, combined with the power quality problem, focusing on the voltage regulation of distribution networks containing distributed PV with a high penetration rate in order to study the solution to ensure the reliable operation of the power system. Currently, the configuration of the energy storage system [3], reactive power regulation of PV inverter [4], active power reduction of PV [5], and tap action of the on-load regulator transformer [6] become the key solutions to the problem of voltage overrun of distribution network with high penetration of PV.

This review firstly summarizes and combs the voltage regulation methods for LV distribution networks containing high penetration distributed PV at home and abroad, firstly analyzes the voltage control methods for LV distribution networks based on energy storage system configuration, reactive power regulation, active power curtailment, OLTC, and distribution network reconfiguration, and then reviews the characteristics, adaptive scenarios, strengths, and deficiencies of the four types of commonly used voltage control methods, and then presents them in the form of tables. After that, voltage control methods coordinated by multiple devices are analyzed; then voltage control is considered from the perspective of flexibility resources, and the important role of load-side resources in voltage regulation is analyzed from three parts focusing on the important role of load-side flexibility resources in voltage regulation, the development trend of load-side resources, and the suggestions to promote the coordination of source-grid-load-storage flexibility resources, which have become one of the main regulation resources of the new type of power system, aiming to Load-side resources have become one of the main regulating resources in the new power system, and it aims to promote the application of load-side resources in voltage regulation in low-voltage distribution networks. Challenges to the development of voltage control in LV distribution networks are also analyzed for the reference of related researchers.

2. Low-Voltage Distribution Network Voltage Regulation Methods and Comparative Analysis

2.1. Energy Storage System Configuration

The working principle of this method is to utilize the energy storage system to store electricity when the load demand is less than the distributed PV generation output, to avoid reverse power flow at the point of common coupling (PCC), and to suppress overvoltage.

Common energy storage systems include flywheel energy storage, pumped storage, electrochemical energy storage, etc. [7]. Electrochemical energy storage is currently the main choice due to its advantages of fast response time and flexible configuration. The PV energy storage access system topology is shown in Figure 1.

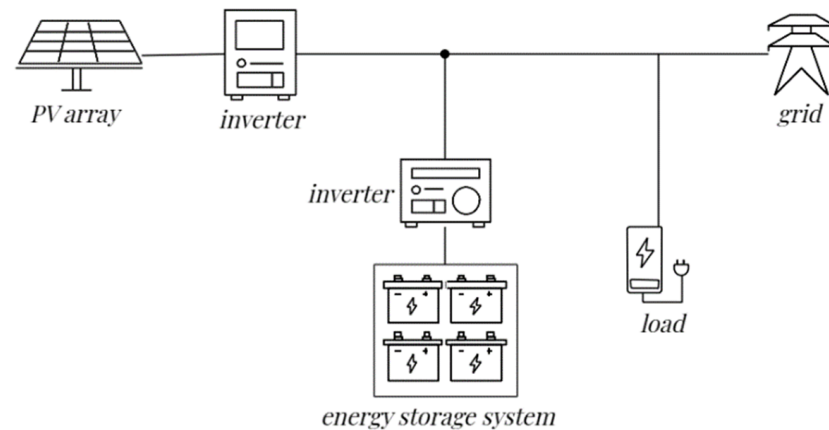


Figure 1. Schematic diagram of a typical PV energy storage access system.

Cheng et al. [8] propose a two-layer coordinated planning model for the energy storage system and its communication network (CN), considering the coupling effect of CPS (Cyber-Physical Systems) to solve the voltage overrun problem. Zoning planning for PV energy storage and constructing a two-layer coordinated planning model can solve the problem of mismatch between load demand and PV output time and improve the voltage stability of the distribution network [9]. Considering the two-layer nested model for distributed energy storage (DES) planning, it can effectively improve the voltage security margin and reduce the subsequent voltage management cost [10]. Zhang et al. [11] propose a source-network-load-storage hierarchical coordinated optimal control strategy to cope with the distribution grid scheduling problem caused by the county-wide PV grid integration, which optimizes the energy storage measurement and the load side through the algorithm to improve the convergence speed and the economic operation efficiency of the grid. The two-tier planning model considers the energy storage system to have a better economy and be more robust.

Chen et al. [12] propose a segmented compensation strategy to reduce the charging and discharging frequency of the energy storage system while smoothing the power fluctuation of distributed photovoltaic power generation, which improves the stability and lifetime of the energy storage system. The study of the capacity, location, and scheduling of distributed energy storage in unbalanced distribution networks can better cope with the uncertainties involved in generating power sources and loads and effectively stabilize the distribution network voltage [13,14]. Distributed control of the two stages of PV inverter and energy storage is currently being applied more widely, using the storage charge state as a consistent variable, which takes into account the capacity of the storage and achieves simultaneous control of the storage power and charge state [15]. Nowadays, deep reinforcement learning for optimal maintenance of distribution network voltage has become a research hotspot, which can better match the photovoltaic power output and load demand characteristics by considering the energy storage charge state to extend the storage capacity with the established objectives and constraints [16].

Advantages of this method:

1. It has a bidirectional regulation function, fast response speed, flexible configuration, and significant economic benefits.
2. It can provide energy for the system during peak power hours, which helps to improve voltage quality and reduce network losses.

3. It can effectively stabilize the fluctuation of distributed photovoltaic power generation so that the photovoltaic power generation system is transformed from an uncontrollable to a controllable power source, which enhances its operability in the distribution network.

Disadvantages of this method:

1. Most of the energy storage models used in the current study are simplified models, which cannot accurately assess the utilization value of the energy storage cycle;
2. a single business model for user-side energy storage, coupled with the frequent occurrence of safety accidents in energy storage, which restricts its application on a large scale;
3. high cost of the energy storage device, short service life, and poor economics for large-scale equipment.

2.2. PV Inverter Reactive Power Regulation

In low voltage distribution networks with large line R/X ratios, reactive power control of PV inverters is an effective means of voltage regulation, which is economically optimal for the control of this scheme compared to controlling PV active, distributed energy storage active, and tapping devices [17].

In recent years, more and more low-voltage household photovoltaic power generation is connected to the grid through inverters with reactive power regulation capability, and the relationship between adjustable reactive power capacity and inverter capacity for [18]

$$Q_{PV}^{max} = \pm \sqrt{S_{IN}^2 - P_{PV}^2} \quad (1)$$

In the formula: Q_{PV}^{max} is the maximum reactive power output capacity of the inverter; P_{PV} is the active power generated by the PV; S_{IN} is the capacity of the inverter, which is about 1.0~1.1 times of the rated active capacity [18].

Figure 2 shows the capacity curve of the PV inverter. p\Point A PV grid-connected active power is the rated power P_{PV}^{rated} . The maximum reactive power output capacity of the inverter is $\pm Q_{PV}^{max1}$. Only by appropriately increasing the capacity of the inverter can the inverter obtain a strong reactive power regulation capability. Meanwhile, with the change of PV grid-connected power, the inverter reactive capacity is in the process of dynamic change, for example, in Figure 2, when the PV grid-connected active power is reduced to βP_{PV}^{rated} ($\beta \in (0,1)$), the inverter operation point is shifted from point A to point B, and the maximum reactive power output capacity of the inverter is increased to $\pm Q_{PV}^{max2}$; at night, the value of the adjustable reactive capacity is equal to the value of the inverter capacity when the grid active output is 0 kW.

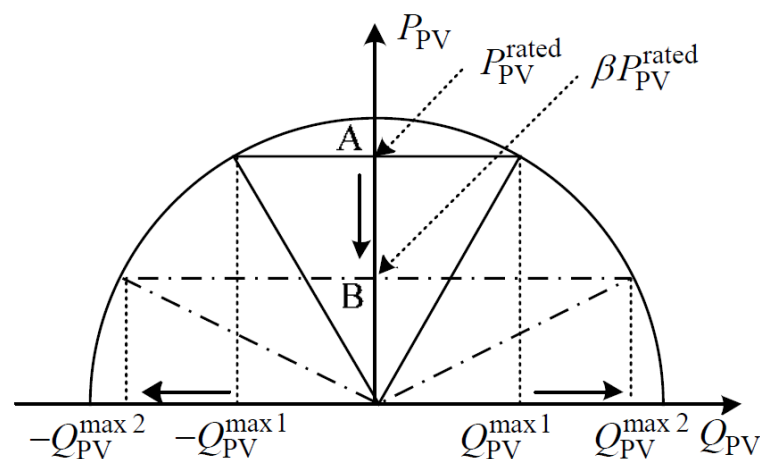


Figure 2. Schematic diagram of the active and reactive capacity of the inverter.

Low-voltage distribution network reactive power control is mainly in situ, and the mainstream strategies can be divided into two [17,19]: $\cos\varphi$ (P) control (with PV grid-connected active P_{PV} as the control input, adjusting the inverter reactive power to control the grid-connected power factor of the PV inverters, $\cos\varphi$), and $Q(U)$ control (with the PV grid-connected voltage magnitude, V , as the control input to achieve inverter reactive power Q_{PV} regulation).

The method works on the principle that when the reverse power flow causes voltage overruns, the distributed PV autonomously regulates its reactive power output, thus keeping the node voltage in the normal operating range.

(1) $Q(U)$ control based on voltage magnitude at the grid point

$Q(U)$ sag control is a classical voltage control method [20–22], and the control curve is shown in Figure 3. When the PV grid-connected point voltage is higher than the target voltage, the PV inverter absorbs reactive power to slow down the rise of the node voltage; conversely, the PV inverter injects reactive power to slow down the fall of the node voltage; when the grid-connected point voltage reaches the upper (lower) limit of the network voltage, the PV inverter absorbs (injects) reactive power according to the maximum reactive power capacity.

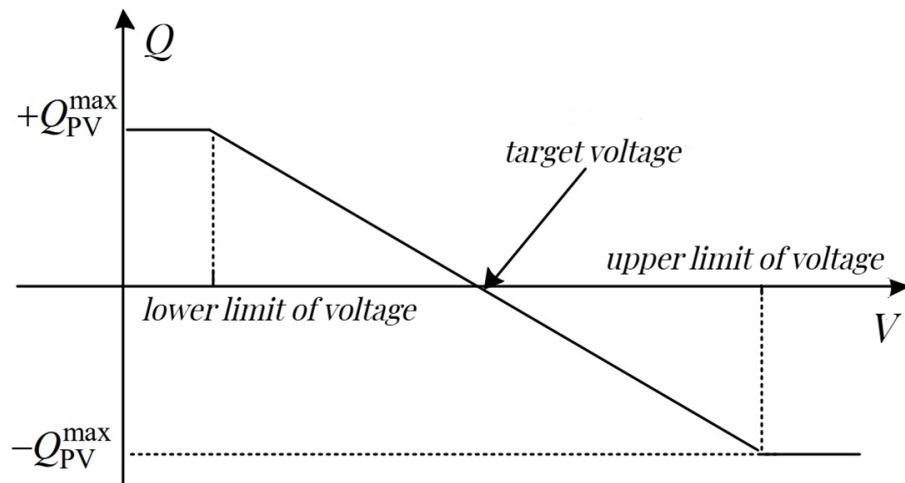


Figure 3. $Q(U)$ control curve.

(2) Active power output based $\cos\varphi$ (P) control

The $\cos\varphi$ (P) control is a typical reactive power control method to limit the network voltage lift and excessive reactive power flow in the line, and the German PV grid-connected standards committee proposed a control curve for distributed PV grid-connected [23], as shown in Figure 4. When the PV grid-connected power exceeds 50% of the rated active power, the PV inverter absorbs the reactive power to reduce the network voltage rise, and at the same time, it should ensure that the power factor at the PV grid-connected point is maintained within the range of ± 0.95 . China's PV power plant grid connection standard also stipulates that the power factor of the PV grid connection point should be maintained within the range of ± 0.95 [24]. This control method in the low-voltage distribution network containing a high proportion of household PV, with strict limitations of the grid-connected power factor of PV, cannot give full play to the reactive power regulation capability of the inverter.

The current study usually divides the grid-connected PV power generation system into three scenarios: over-voltage suppression, under-voltage suppression, and optimization of network loss and power factor in order to make full use of the reactive voltage control potential of the inverter [25]. By considering the active distribution network voltage reactive power coordinated optimization control strategy under different voltage levels [26],

the reactive power output is adjusted in real-time to suppress voltage overruns and fluctuations [27]. Liu et al. [28] proposed a multi-timescale coordinated control method based on reactive power/voltage optimization at the point of photovoltaic co-coupling (PCC), which can adaptively regulate the reactive voltage of the PCC according to the real-time variations of the PV and load outputs. Considering distributed coordinated voltage control for PV storage, the reactive power of the PV inverter and active power regulation capability of the storage system are reasonably utilized to avoid the regulation limitation problem brought by single link control [29].

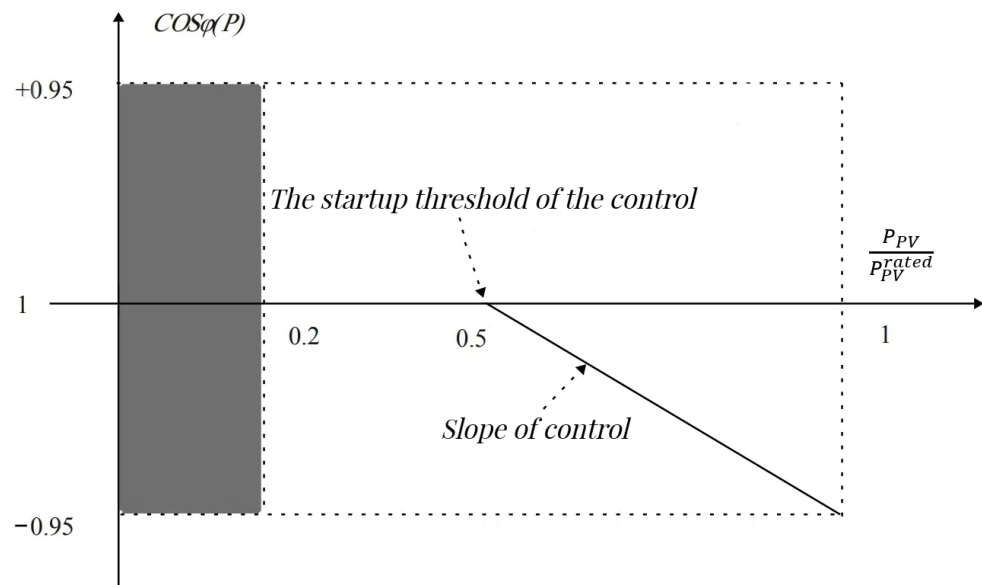


Figure 4. $\cos \varphi (P)$ control curve.

Simultaneous consideration of PV inverter reactive capacity regulation and reduction of PV active curtailment are the commonly used voltage control methods. Optimizing the reactive power capacity of smart PV inverters through a centralized active and reactive power management system (CARPMS) can alleviate the voltage overrun problem [30]. Utilizing the reactive power capacity of inverters to regulate the system node voltage can also effectively reduce the system network loss and PV active curtailment and largely eliminate the impact of distributed renewable energy fluctuations on the system voltage in the distribution network [31].

Liu et al. [32] propose a power control strategy based on the voltage magnitude and line impedance at the grid-connection point, which utilizes the residual capacity of the inverter when the voltage at the grid-connection point exceeds the upper limit to minimize PV abandonment under the premise of ensuring voltage quality. Considering the capacity characteristics of PV inverters and node voltage offset limitations, voltage control using inverter reactive power can reduce the PV grid-connected active curtailment and, at the same time, achieve the optimization of network losses [33]. Li et al. [34] propose an adaptive two-layer stochastic approach that introduces a Volt/VAR control (VVC) scheme to ensure voltage safety and efficient operation and a conditional value-at-risk (CVaR) based risk assessment method. The method transforms the original nonlinear problem into a mixed-integer linear programming (MILP) model, which ensures high-quality solutions and computational performance and can effectively ensure system safety and reduce operational costs and risks.

Zhang et al. [35] address the reactive power output of PV power supply seriously threatening the reliable operation of PV inverter and proposes a data-driven voltage reactive power optimization control strategy considering the reliability of PV inverter, which utilizes the deep deterministic policy gradient algorithm (DDPG) to realize the voltage reactive power control, and this method introduces reinforcement learning into voltage

reactive power control of the distribution network so that the PV power supply participates in the reactive power regulation of the distribution network has become a current research hotspot.

Advantages of this method:

1. Full utilization of PV power generation as well as the reactive power output capacity of PV inverters;
2. Compared with the traditional centralized voltage regulation method, this method belongs to the locally distributed control with fast response speed.

Disadvantages of this method:

1. transmission lines flow through a large amount of reactive current will increase network losses;
2. reduce the feeder power factor, affecting the quality of power to the user;
3. in the PV power generation rated active power conditions, its reactive power output capacity is limited.

The methodology of voltage regulation for Energy storage system configuration and inverter reactive power regulation section is summarized as shown in Table 1, which includes the research methodology, concerns, and resources considered.

Table 1. Review of related work: Energy storage system configuration and inverter reactive power regulation part.

References	Research Method	Focus	Utilized Resources
[8–10]	Two-tier planning model	voltage stability	PV, energy storage
[11]	Layered coordination optimization	Distribution Network Dispatch	Energy storage, load
[12–14]	Segmented compensation strategy	Voltage stability, Reduced power fluctuations	PV, energy storage
[15]	Two-stage distributed control	energy storage charge state	PV, energy storage
[16]	deep reinforcement learning	voltage stability	PV, energy storage
[25–27]	Delineate optimization scenarios, Distinguish voltage levels	Suppressing voltage overruns	PV
[28]	Coordinated control on multiple time scales	Adjustment of reactive voltage	PV, load
[29]	Distributed coordinated voltage control	Solving the problem of limited control in a single link	PV, energy storage
[30,31]	CARPMS	Suppressing voltage overruns	PV reactive capacity, Active reduction
[32,33]	Regulation using inverter capacity characteristics	Reducing light abandonment, Optimize network losses	Photovoltaic reactive capacity
[34]	Adaptive two-layer stochastic approach, VVC	reduce costs	PV
[35]	Reinforcement learning: DDPG	Data Driver Voltage Control	PV

2.3. PV Active Power Reduction

The working principle of this method is to regulate the voltage by limiting the active power of distributed PV. In low-voltage distribution networks, the method of controlling reactive power to regulate voltage is less effective due to the large distribution network line impedance ratio R/X. Controlling the active power output from distributed PV can provide better voltage regulation.

As shown in Figure 5. represents the relationship between output active and reactive power and voltage when local sag control is used for distributed PV. In the figure, U_n denotes the rated voltage of the node, and U_{UV} and U_{OV} denote the corresponding voltage thresholds when the node has the risk of undervoltage and overvoltage, respectively. When

the voltage at the grid point is lower than U_{OV} , the distributed PV is running in MPPT mode, and when the voltage at the grid point is higher than U_{OV} , the distributed PV reduces the active power, and P_{cut} represents the active power reduction.

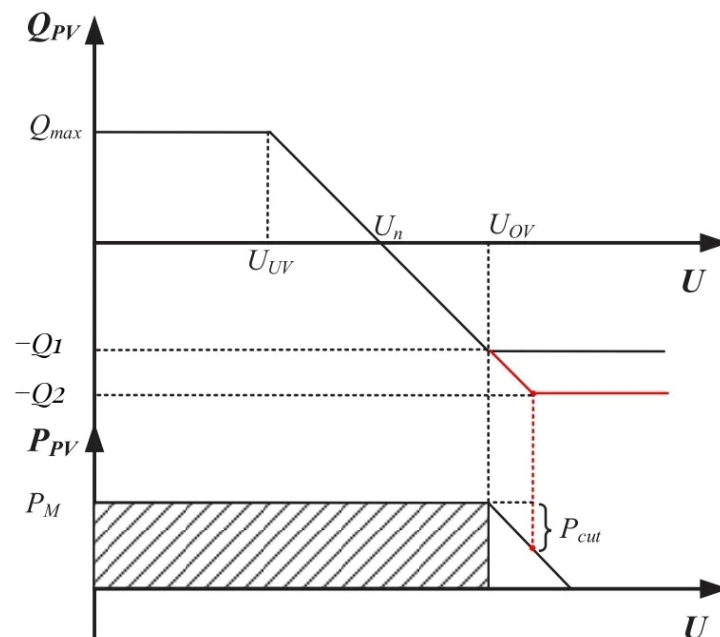


Figure 5. Active and reactive power curves for distributed PV local control.

Reducing PV active power output through a voltage control strategy can realize the effect of suppressing overvoltage, but only reducing PV active power output can not improve the PV consumption-ability of the distribution network [36]. The currently used method is to combine reactive power regulation and active power reduction, which can provide a better voltage regulation effect. For example, Jameel et al. [30] propose a centralized active and reactive power management system (CARPMS) to optimally utilize the reactive power capacity of smart PV inverters and to control the amount of active power curtailment in order to mitigate the voltage overrun problem. When the reactive power capacity of the inverter is insufficient, a portion of the active power output from the PV is curtailed according to priority to meet the reactive power demand of the system. At night or when the system's reactive power deficit is more severe, the PV inverter operates as a static synchronous compensator [37].

Voltage control through in situ adaptive, considering active reduction and reactive power regulation, can provide a better voltage regulation effect in real LV distribution networks [38]. Utilizing smart inverters of PV systems to compensate reactive power in real-time can reduce active curtailment and voltage distortion and reduce the requirements on communication infrastructure as an advantage of voltage regulation [39].

Ma et al. [40] stabilize the output by improving the constant power control strategy so that the active power curtailment control works even when the light intensity changes drastically. Considering accurate prediction of the active power limit, real-time modulation of the active power output of PV arrays can reduce the probability of voltage overruns occurring [41].

When there is a risk of voltage overrun in the distribution network, the PV power supply sets its active power limit value injected into the distribution network in real-time, and the excess PV power is temporarily stored by relying on the energy storage equipment, which can prevent voltage overrun fundamentally, and utilize the residual capacity of the inverter to absorb the inductive reactive power and reduce the voltage of the distribution network [42]. Considering the control strategy combining active curtailment and on-load tap changer to solve the voltage overrun problem becomes a new research idea [43].

Advantages of this method:

1. As the distribution network has the characteristic of larger R/X, the active power reduction has better regulation of overvoltage.
2. This method can reduce the uneven distribution of current and voltage and reduce the line loss and grid loss; there is no problem with network loss increase, feeder power factor reduction, or line overload caused by reactive power regulation.

Disadvantages of this method:

1. An unreasonable selection of parameters will lead to an increase in the amount of active power reduction and losses.
2. Active power reduction will affect the economy of distributed PV.
3. The energy utilization efficiency of the PV system will be reduced, and the PV consumption capacity will be reduced.

2.4. On-Load Regulator Transformer Tap Action and Distribution Network Reconfiguration

As shown in Figure 6, the principle of the tap adjustment method of the on-load regulator transformer is to realize the voltage adjustment by adjusting the tap position of the high-voltage winding to adapt to the changes of the incoming power supply and the electrical load in the operation of the system when the voltage overrun problem occurs in the line [44]. The purpose of adjusting the output voltage of the low-voltage winding at any time is realized by switching the tap switch at any time and adjusting the tap position of the high-voltage winding under the condition of carrying load to ensure the quality of the output voltage.

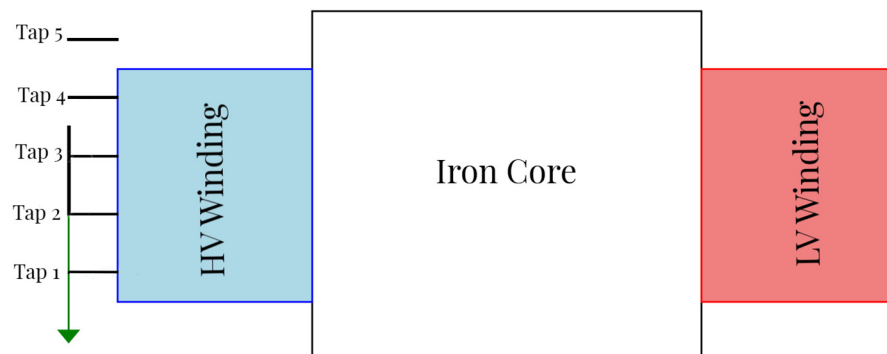


Figure 6. On-load Tap Changer Operation in Transformer. The green arrow refers to the meaning of the transformer high voltage side tap gradually decreasing from high to low, and is distinguished from the black diagram of the five taps.

Yuan et al. [45] analyze the principle of the double-core symmetrical phase-shifting transformer (DS-PST) and the application of trend regulation, which can transfer the active power transmitted by high load-rate lines to low load-rate lines so as to make the distribution of system trend more balanced and effectively improve the transmission capacity of the power system. Changing the control strategy of the on-load tap changer (OLTC), the first controller (ANFIS-OC) in this strategy solves the voltage overrun problem due to low X/R ratio and PV distributed generation by calculating the OLTC reference voltage based on the minimum and maximum voltage magnitude of the grid [46]. Tang et al. [47] innovate the voltage control architecture for distribution networks by considering the modeling of the discrete control of on-load tap-changer transformers, which is applicable to the coordinated control of a fundamentally developed network, as well as to the optimization-based centralized control of a network with a fully articulated system.

Since the traditional OLTC for control means can no longer resist the impact of high penetration distributed power sources on the grid, the dynamic reconfiguration of distribution networks containing distributed power sources is investigated. Transformer economic

operation as a measure of grid energy saving and consumption reduction, distribution network reconfiguration, distributed power reactive power coordination, and optimization can make the distribution network a wider range of improved energy efficiency.

The integrated use of real-time phase measurement unit (PMU) data to coordinate the optimal control of the battery energy storage system (BESS), OLTC, and solar PV inverters, reduce the OLTC tapping operation, and increase the service life of the OLTC and BESS is a feasible solution to improve the system performance [48]. Coordinated voltage regulation by regulating the reactive power of inverter-type distributed power supply (DG) and OLTC can reduce the operational burden of OLTC, using curve fitting techniques to achieve stable regulation within the voltage range, solving the voltage overrun problem in the distribution network, and providing a feasible way to improve the performance of the distribution system [49].

In recent years, artificial intelligence algorithms based on randomization techniques have been widely used in the field of optimization. Among the examples of intelligent optimization algorithms applied to distribution network reconfiguration, it is summarized that they mainly include artificial neural network algorithm, simulated annealing algorithm, forbidden search algorithm, ant colony algorithm, particle swarm optimization algorithm, and genetic algorithm, differential evolutionary algorithm, etc., and the use of a new type of hybrid algorithms to carry out the reconfiguration of the distribution network becomes a hot spot of today's research [50].

Cui et al. [51] proposed a PV admittance capacity improvement model based on network reconfiguration and reactive voltage regulation. Controllable distributed power sources, OLTCs, castable capacitors, and distribution network reconfiguration are used as regulation objects to effectively improve the PV admittance capacity of the distribution system through distribution network reconfiguration and reactive voltage regulation. The coordinated optimization of OLTCs, distribution static compensators (D-STATCOM), distribution generation (DG), and distribution network reconfiguration can maximize the annual cost savings and improve the voltage profile [52].

Advantages of this method:

1. Adjusting the tap switch position of the on-load voltage regulator allows the transformer to be carried out in real time, and the system can respond quickly to voltage overruns caused by fluctuations in PV power, thus improving the stability of the power grid.
2. The adjustment of the tap switch position of the transformer can be adapted to the different operating conditions and the changes in the output of PV power, thus making the system more flexible.
3. It is economical and does not require additional equipment to realize voltage control.
4. Transformer regulation can be combined with distribution network reconfiguration to improve the efficiency of energy utilization.

Disadvantages of this method:

1. Adjustment of the switch position of the on-load regulator transformer will lead to transformer losses and current and magnetic losses will be generated during the adjustment process, affecting the overall efficiency of the system.
2. Although it is an instantaneous adjustment, the response speed of the on-load regulator transformer is relatively slow.
3. Frequent adjustment of the switch position of the transformer taps affects the life of the equipment, and it is necessary to consider the durability of the equipment and the cost of maintenance.

The methodology of PV active power reduction and OLTC and distribution network reconfiguration section is summarized as shown in Table 2, which includes the research methodology, concerns, and resources considered.

Table 2. Review of related work: PV active power reduction and OLTC and distribution network reconfiguration section.

References	Research Method	Focus	Utilized Resources
[36,37]	Determine active cuts based on prioritization, Inverter as static synchronous compensator	Meet system reactive power requirements	PV, active reduction
[38,39]	Local adaptive voltage control	Reduced active cuts, Reduced communication requirements	PV, active reduction
[40,41]	Power prediction methods	voltage stability	PV, active reduction
[42]	Utilization of energy storage systems to store electrical energy	Preventing voltage overruns	PV, energy storage, active reduction
[43]	OLTC combined with active reduction	voltage stability	PV, active reduction, OLTC
[45]	trend regulation	Enhancement of power system transmission capacity	PV, OLTC
[46]	Improved control strategies	Preventing voltage overruns	PV, OLTC
[47]	Distributed coordinated voltage control	Solving the problem of limited control in a single link	PV, energy storage
[48,49]	Coordinate multiple resources with PMU Optimize control	Enhances system performance, Voltage Stability	PV, Energy Storage, OLTC
[51,52]	Regulation of multiple resources, Modeling PV Consumption Capacity Enhancement	Cost savings, Improved voltage	PV, OLTC, network reconfiguration

2.5. Comparative Analysis of Methods

According to the previous section, it can be seen that the above four types of methods are able to solve the voltage overrun problem caused by high penetration distributed PV access, but each method has its own advantages and limitations, and the comparative analysis is shown in Table 3.

Table 3. Comparative analysis of four types of voltage regulation methods.

	Energy Storage System Configuration	PV Inverter Reactive Power Regulation	PV Active Power Reduction	OLTC and Distribution Network Reconfiguration
Implementation costs	high	low	relatively low	low
Responsiveness	quick	quick	quick	slow
Adjustment performance	excellent	excellent	relatively excellent	poor
Network loss	minimize	rise	minimize	invariant
Economic gain	high	low	relatively low	high
Adjusting the difficulty	relatively hard	easy	easy	difficult

In summary, when solving the voltage overrun problem in the distribution network, a single voltage regulation method has limited effect and is insufficient to meet the grid connection requirements of high penetration distributed PV in most cases. Combining the advantages and disadvantages of each voltage regulation method, research scholars have proposed a coordinated control strategy with multiple methods.

For example, Zhang et al. [53] propose a coordinated voltage control strategy for OLTC and energy storage system, which utilizes OLTC and energy storage system to achieve

coarse and fine regulation of distribution network voltage, respectively, and effectively suppresses voltage overruns through two-phase operation optimization with different time scales and regulation ranges.

Tang et al. [54] analyze that the inverter-based reactive power control strategy has limited voltage regulation capability, resulting in network loss and power factor index degradation, while the storage-based active control strategy can relatively effectively regulate the voltage and optimize the network loss but a single active control has a greater demand for storage capacity, which is not conducive to optimizing the network power factor. Based on this, a coordinated control strategy of inverter reactive power and storage active power is proposed, which can effectively regulate the network voltage, comprehensively optimize the technical indexes of the grid, and reduce the requirement of energy storage capacity.

Li et al. [42] propose an integrated control strategy of PV active reduction and inverter reactive power. In case of voltage overrun risk in the distribution network, the PV power supply sets its active power limit value injected into the distribution network in real-time to prevent voltage overrun fundamentally and, at the same time, utilizes the residual capacity of the inverter to absorb inductive reactive power in order to further reduce the voltage of the distribution network and avoid unnecessary active reduction.

As shown in Figure 7, it can be seen that the mainstream method of solving the voltage overrun problem in LV distribution networks today is to coordinate the control strategy for voltage regulation through multiple methods, but there are still some problems to be considered under such methods:

- (1) A PV grid-connected low voltage distribution network with a high proportion of poor communication and measurement conditions affects the coordination and control capability of the distribution network itself, often requiring the simultaneous use of PV inverters, energy storage, OLTC, and other devices to suppress voltage overruns and voltage fluctuations. When multiple devices are regulated at the same time, there is a large difference in the regulation effect achieved by different regulation sequences; while the economic cost of a single regulation of OLTC is high, the degree of coordination of the source network and load needs to be improved, and the research on the control model that considers the economic factors are still relatively small, and the regulation sequences of the individual devices as well as the regulation economy need to be further investigated.
- (2) The control objectives and scenarios of most of the current control methods are relatively single, i.e., the control does not fully consider the operating scenarios of the PV, and there is the problem of insufficient consideration of the network operation index. Nowadays, a large number of new elements are integrated into the load side, the load resources are more complex and variable, and it is more difficult to obtain the basic data of the scaled flexible resources. However, in terms of data acquisition, the load usually adopts probabilistic statistical data rather than measured data, which is not representative of today's changing load resources; in terms of model construction, simplified models are usually adopted, and the dynamic characteristics are poorly displayed.
- (3) Compared with a low-voltage AC distribution network, a low-voltage DC distribution network has the characteristics of small line loss, low cost, etc., household photovoltaic and energy storage is also easier to access, and at the same time, has better power supply reliability and power quality. The transition from low-voltage AC distribution networks to low-voltage AC/DC hybrid distribution networks has become one of the current trends in network development. The application of some new power electronic equipment in low-voltage distribution networks is receiving attention, and the application of new equipment in low-voltage distribution networks has to be supplemented and improved.
- (4) Policy factors are the main driving force in promoting distributed household PV grid connections, but they will eventually be regulated by the market mechanism. Based

on certain price elements, the PV grid connection network voltage control and other issues will be more reasonable.

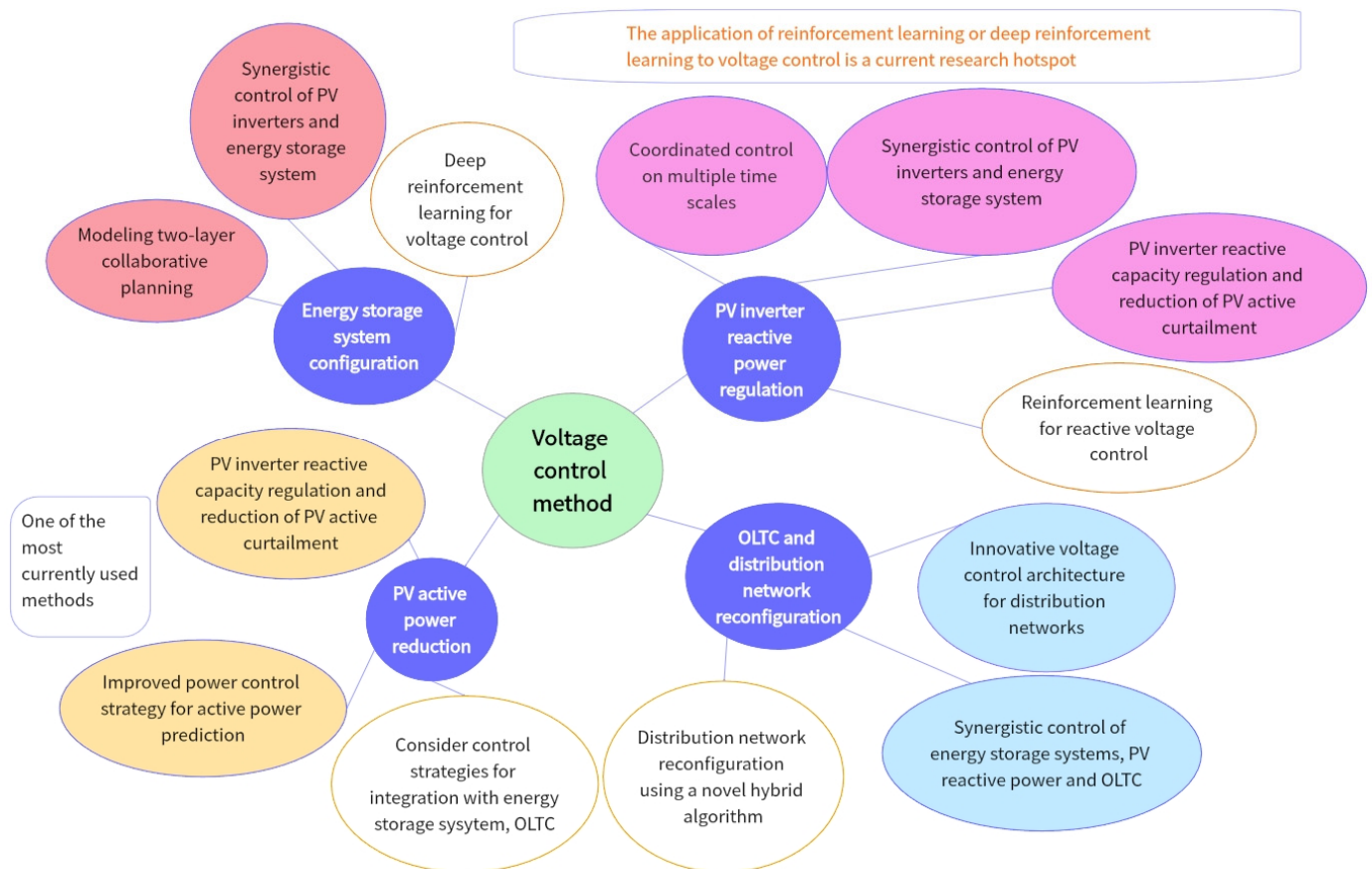


Figure 7. Voltage regulation methods and analysis in LV distribution networks.

3. Flexibility of LV Distribution Networks Resource Voltage Regulation

With the large-scale grid connection of distributed PV, the source and load ends of the power system show a high degree of uncertainty, the stable operation mechanism of the power system is more diversified and complex, and the overall characteristics of the high proportion of new energy power system have changed dramatically. The high penetration rate of a high proportion of distributed PV leads to the lack of flexibility of the system in local time, and it is difficult to guarantee the safe, reliable, and economic operation of the new power system by relying only on the power side regulation. Demand-side resources have the characteristics of wide distribution, small single capacity, and diversified means of regulation, etc. The formation of the cluster effect can promote the development of a power system from “source follows load” to “source-load interaction”, which can further satisfy the demand for distributed power consumption.

The voltage regulation methods for high PV penetration in LV distribution networks in the previous section have been introduced for the three types of flexibility resources of source, network, and storage, i.e., considering the distributed PV itself for voltage regulation, OLTC and distribution network reconfiguration for voltage regulation, and energy storage system configuration for voltage regulation. Through comparative analysis, it can be seen that the three types of flexibility resources, namely source, network, and storage, have been widely used in distribution network voltage regulation, but there are insufficient advantages and obvious defects in using one method alone. While analyzing from the perspective of flexibility resources, load-side resources have become one of the main regulating resources of the new power system, but the current study introduces less about the participation of load-side flexibility resources in voltage regulation of LV

distribution networks and advancing the application of load-side resources in voltage regulation of LV distribution networks is the focus of future research. The following section focuses on the analysis of load-side flexibility resources and describes the important role of load-side resources in voltage regulation from three parts: the important role of load-side resources, the development trend of load-side resources, and the proposal to promote the coordination of source-network-load-storage flexibility resources.

3.1. Load-Side Flexibility Resource Voltage Regulation

Domestic and international views on flexibility resources are basically consistent, with conventional power plants, energy storage, interconnected grids, and the demand side being the main components of power system flexibility resources. However, the rapid expansion of large-scale renewable energy grid connections and various end-use electrification makes the power system pay more attention to the load-side flexibility, that is, the demand-side flexibility. According to the experience of European power grid operation and scheduling, fully utilizing the flexibility potential of demand-side resources can solve the net load fluctuation problem caused by distributed power supply access and other factors, make up for the lack of flexibility of the distribution network, and significantly improve the economy, reliability, and flexibility of power system operation.

Demand-side flexibility resources mainly include adjustable loads, electric vehicles, user-side energy storage, and other small and decentralized “producers and consumers”, and with the continuous improvement of user-side intelligence and automation, demand-side resources can give full play to their flexible and controllable potential to a greater extent. Load-side resources have the ability to actively respond to system power fluctuations, and controllable loads, as one of the means of demand-side management, can provide the system with demand-side flexibility resources, and their rapid response capability can meet the requirements of system load demand changes. However, due to the problems of scattered demand-side resources, large differences in user energy use, and small size of adjustable load, it is difficult for demand-side flexible resources to directly participate in the centralized market, provide auxiliary services through aggregator agents and other forms, and realize the unified management of internal decentralized resources by means of advanced communication technology in order to realize the voltage regulation function.

The literature [55] proposes a game model between the distribution system and load aggregator, where both the distribution system and load aggregator can obtain optimal economic benefits and alleviate the pressure of distribution network voltage regulation through demand response.

Considering the reactive power regulation capability of PV inverters and active distribution voltage regulation based on demand response flexibility, the coordination between PV inverters and demand response can be fully utilized to improve the power quality and reliability of the distribution network [56]. In the LV distribution network system with high penetration of PV power generation and electric vehicles, the bidirectional overrun problem leads to the difficulty of realizing voltage regulation, and the utilization of flexible loads through demand response can effectively carry out voltage regulation and reduce the voltage overrun problem [57].

In [58], a two-layer optimization model for the coordinated allocation of multiple types of reactive power compensation resources based on a “planning-operation” integration framework is proposed for the reactive power voltage problem caused by large-scale distributed new energy access. The framework takes into account the relationship between “source and load”, and aims to improve voltage quality, consume new energy, and reduce the impact of power back-feeding to adapt to the future development trend of the power distribution system.

Power supply, interconnected grid, flexible loads, and energy storage in the power system can provide a certain degree of flexibility [59]. Under a high proportion of new energy access, there is the problem of increased volatility and stochasticity on the supply side of the power system, and the system shortages and abandonment rates also increase

the use of energy storage systems and interconnected grids to carry out the optimal capacity planning, which can satisfy the demand for system flexibility [60]. Establishing a collaborative planning model for source-load-storage flexibility resources is a prioritized option. An expected unserved ramping (EUR) metric is proposed, which can indicate the amount of flexibility resources that still need to be invoked to meet the system demand for further collaborative planning [61]. The collaborative planning model of source-load-storage flexibility resources can meet the needs of distribution network power supply and consumption by establishing a collaborative planning model [62]. The scheme can fully consider the supply characteristics and network constraints of various types of resources, realize the unified and coordinated planning of multiple flexibility resources, and optimize the economy and flexibility of the power system [63].

Azarnia et al. [64] considered the dependency between power load and distribution voltage and proposed a Robust Optimization (RO) model to reduce the occurrence of voltage overruns by using Smart Grid technology, Voltage Reactive Control (VVC) unit scheduling system equipment in real-time to keep the voltage in the permissible range so that the cost and voltage overruns can be reduced by decreasing the computational error.

In summary, with the accelerated construction of the new power system, only considering the use of the power supply side, grid side, energy storage measurement of the flexibility of resources has been difficult to meet the system's safe and efficient operation needs, it is necessary to release the flexibility of the demand-side resources regulating capacity, "source-load interaction" will become a significant feature of the future power system. Currently, the utilization of demand-side resources is a major hotspot, and the form of utilization has gradually evolved from a single orderly power consumption and energy efficiency management to a variety of functions, active participation, and market-driven demand response resources. Considering the full utilization of load-side flexibility resources in distribution network voltage regulation fits the future development trend of the power system, further improves the flexibility of the distribution network, and plays a certain effect on the voltage regulation and mitigation of voltage overrun.

3.2. Voltage Regulation of Source, Network, Load, and Storage Flexibility Resources

Each type of flexibility resource in the LV distribution network has its own advantages and limitations. Focusing on technical characteristics and economics, the cost advantages of considering demand response and distributed PV's own regulation capability are obvious, but the regulation capability is slightly insufficient, e.g., considering distributed PV itself may result in active power curtailment, while energy storage system configurations have an advantage in regulating the voltage capability, but they are more expensive. The advantage of considering multiple flexibility resources for voltage regulation is that demand-side response and energy storage systems, etc., share the spare capacity of the new power system under a high proportion of new energy access to meet the power quality requirements under high penetration of new energy and PV consumption requirements.

However, there are the following problems: the construction cost is slightly higher than that of the planning scheme without flexibility constraints; the volatility of the new energy output on the source side will lead to the dual problems of power supply shortage and new energy abandonment; the regulation effect of considering only a single flexibility resource is poor, i.e., the degree of coordination among the source, network, load, and storage is still to be improved. For example, considering only load-side flexibility resources, the voltage regulation range is small, and it is difficult to solve the problem of voltage overrun in both directions.

Further improving the degree of coordination of source, network, load, and storage flexibility resources, as well as scientifically and reasonably playing the role of demand-side resources, have become the focus of attention for the high-quality development of electric power in the medium and long term. The following section will discuss the important role of load-side resources, the development trend of load-side resources, and the proposal to promote the coordination of source, network, load, and storage flexibility resources.

3.2.1. The Important Role of Load-Side Resources in New Power Systems

(1) Ease the contradiction between power supply and demand. With the acceleration of the demand side development process, especially the deep integration of the power network and transportation network, the difficulty of balancing supply and demand in the power system gradually increases, and short-term load spikes occur frequently. Due to the low substitution effect of photovoltaic capacity, under the trend of new energy as the main source of power increment, the active play of load-side resources in power peak shaving and valley filling will help to guarantee the balance of power supply and demand.

(2) Promote new energy consumption. PV power generation has strong randomness, volatility, and intermittency. In the context of large-scale access to the grid by new energy sources, relying solely on new power-side regulation resources to support new energy consumption will face problems such as lower equipment utilization efficiency and higher electricity costs. Load-side resources can respond up to the second level, with excellent regulation capabilities, such as electric vehicles, user-side energy storage, central HVAC and other demand-side resources can quickly respond to new energy ultra-short, short-cycle scale regulation needs; traditional industrial and commercial loads and other load-side resources can be based on incentives to actively participate in the system within the day regulation needs; hydrogen and other emerging load-side resources and the new energy depth of the coupling can meet the new energy multi-day or longer time scales, the new energy or longer time scales. Emerging load-side resources, such as hydrogen energy, are deeply coupled with new energy to meet the multi-day or longer time-scale regulation demand of new energy [65].

(3) Enhance the energy efficiency level of the whole society. Load-side resources are important intrinsic resources for the power system to tap its own energy-saving potential. According to the classification of saving effect, they can be divided into adjustable power resources and saving power resources, the former through load transfer, load regulation, load interruption, and other regulation methods to replace or reduce the power supply, power grid type of construction investment, and the latter through process optimization, technological improvement, management enhancement, and other means to save demand-side power and reduce the level of energy consumption.

3.2.2. Trends in Future Development of Load-Side Resources

(1) Load-side resources will become one of the main regulating resources of the new power system. With the accelerated construction of the new power system, load-side resources will gradually be elevated to the same status as the supply side, playing an important role in ensuring the balance of power supply and demand and supporting the consumption of new energy. Load-side resources can be adjusted with huge potential, and some traditional industrial and commercial loads will still be the main load-side regulation resources in the future due to their large scale and high electricity price sensitivity; emerging load-side resources such as electric vehicles and customer-side energy storage, which are fast-responsive and highly flexible, will certainly play an increasingly important role [66].

(2) Digitalization and intelligence accelerate the efficient use of load-side resources. At present, big data, cloud computing, blockchain, 5G communication technology, and other fields continue to innovate and breakthroughs, and in the future, with the massive load-side resource access system, load-side resource digitization and intelligence become an inevitable development trend, deepen the data analysis in the fields of user behavior analysis and prediction, resource potential mining, accurate response implementation, etc., and strengthen the resource aggregation and invitation, multi-user interaction, source network, load and storage coordination control and other aspects. The intelligent application provides key support for the scientific utilization of load-side resources [67,68].

(3) Realize the deep integration of multiple energy sources on the load side. With the acceleration of the development process, a new generation of energy use methods such as integrated energy, vehicle-network interaction, microgrids, and virtual power plants are flourishing, and the degree of coupling between multiple energy systems on the load side

has increased. The boundaries of energy producers and energy consumers are gradually blurred, and the power network, heat network, and transportation network to achieve in-depth integration, electricity, heat, hydrogen, etc., to the power system as a hub for mutual conversion and mutual aid, load-side resources will play an important role in promoting the consumption of clean energy and the green and low-carbon transformation of energy [69].

(4) The degree of marketization of load-side resources will be greatly increased. Load-side resources are close to users and have natural marketization conditions. In the future, various types of load-side resources will gradually transition from incentive compensation to active participation in power market-oriented transactions, and the willingness and ability of load-side resources to participate in system regulation will be accelerated. By participating in the electric energy market, auxiliary service market, capacity market, and other trading varieties, the commodity price attribute of load-side resources in the new type of power system has been fully embodied, and the optimization and allocation efficiency of load-side resources has been effectively enhanced [70].

3.2.3. Recommendations for Promoting the Coordination of Source, Network, Load, and Storage Flexibility Resources

(1) Establish a long-term load-side resource incentive mechanism. Following the principle of fairness and reasonableness, establish a long-term incentive mechanism, cultivate the sense of participation of market players, guide large industrial and commercial users to directly participate in demand response through the direct transaction mode of large users, encourage power sales companies and comprehensive energy service companies to take on the role of load aggregators, and promote the model of virtual power plants in cities with more abundant electric vehicle and commercial building load resources. Multi-channel mobilization of incentive funds, the formation of a long-term stable pool of funds, and the gradual transformation of incentive-based demand response from a temporary and emergency initiative to a normalized means of regulation will help to promote the coordinated development of flexibility resources such as load-side resources [71].

(2) Improve the participation of load-side resources in the market mechanism. Enrich the electric power spot market trading varieties, study and establish the load-side resources to participate in the energy market, auxiliary services, capacity market organization mode, establish and improve the market access mechanism and other content, improve the demand response market-oriented trading regulatory system, and promote the formation of the supply-side and demand-side joint participation, fair competition, market-oriented trading model [72].

(3) Promote the technological progress of load-side resources. Vigorously develop automatic demand response technology, load aggregation technology, power saving measurement and verification technology, intelligent power consumption facilities, and intelligent load control technology. Explore the “Internet+” intelligent power consumption technology model and organization model. Through the extensive deployment of data collection terminals for user information, grid information, and power generation information, break the data barriers of source-network-load-storage, integrate system operation, market transaction, and user electricity consumption data, improve the capacity of load-side big data analysis, and realize the intelligent regulation and control of load-side resources [73].

In summary, the flexible adjustability of the load side in the LV distribution network flexibility resources is particularly important. Voltage regulation through source, network, load, and storage flexibility resources is shown in Figure 8. Consumption of PV in the LV distribution network not only needs to be balanced locally but also should consider inter-regional inter-supply and overall coordination of consumption through multi-energy complementary and regional interconnection to improve the capacity of PV consumption so as to dissolve the risk of network voltage overruns. Voltage control in the low-voltage distribution network containing high-penetration distributed PV requires rational planning of flexibility resources, considering the important role of load-side resources, rationally

promoting the further coordinated development of source, network, load, and storage flexibility resources, and fully exploring the flexibility potential of each side of the source-network-load-storage as well as the grid's role in supporting the flexibility of transmission, in order to improve the flexibility and economy of the system [74].

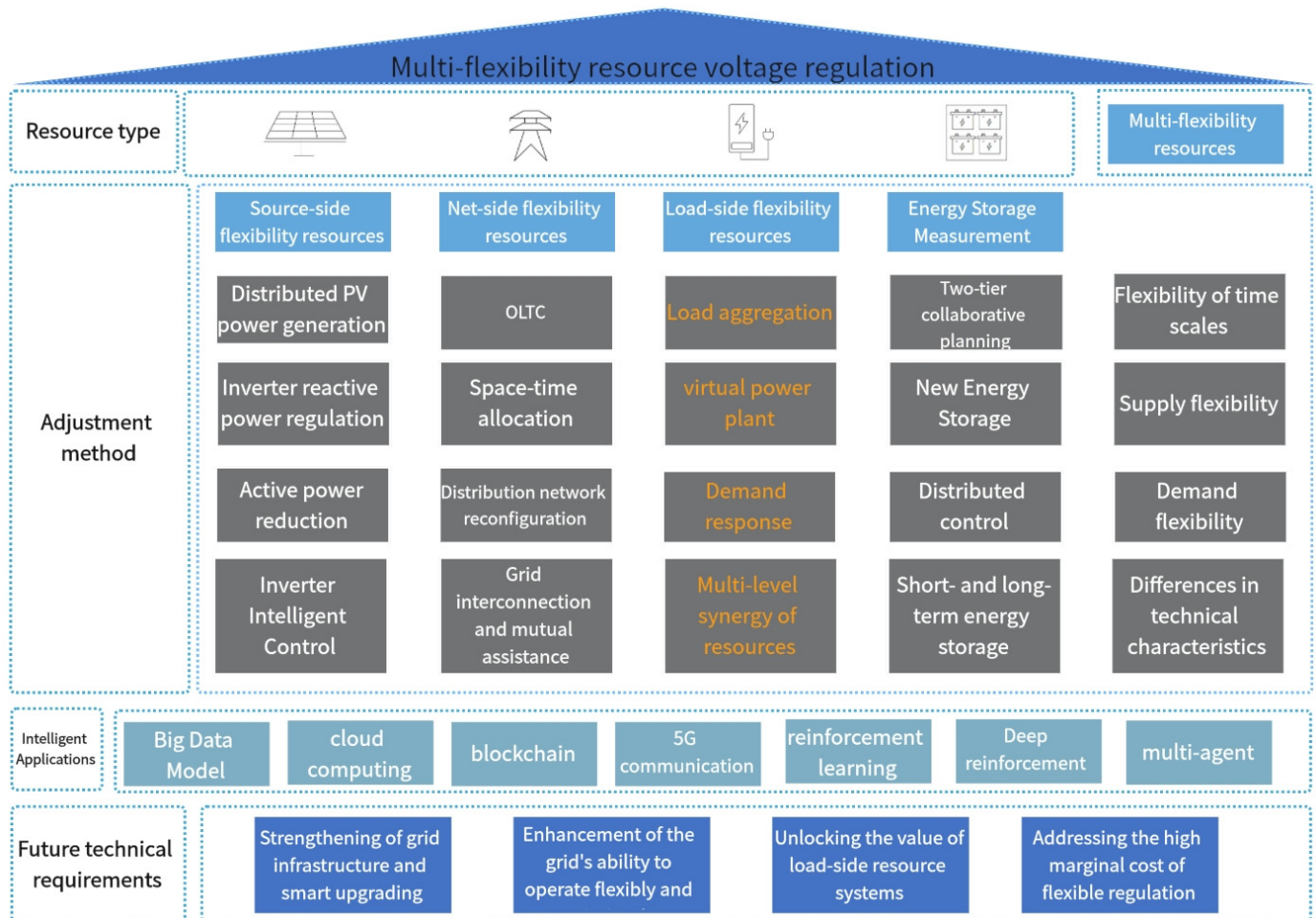


Figure 8. Multi-flexibility resource voltage regulation.

4. Problems and Recommendations for Voltage Control

4.1. Challenges

Distribution grid voltage regulation methods containing high penetration distributed PV and some of the problems in the mainstream methods mentioned above are as follows:

(1) When voltage regulation is carried out using a coordinated control strategy with multiple methods, there is a problem of not considering the regulation sequence and economy of multiple methods, and although the purpose of voltage regulation is achieved, there are problems such as poor economy and low efficiency.

(2) In the process of low voltage distribution network modeling, the existing research methods simplify the data, model, and parameter selection, lack of consideration of PV operation scenarios, and have the problems of insufficient consideration of network operation indexes and oversimplification of data, modeling and parameter selection.

(3) In the face of the sharp increase in electricity consumption of the whole society, subject to environmental protection, geography, and other factors, it is difficult to take the way of large-scale planning and construction of new lines to expand capacity and there is the problem that the existing line structure can not meet the demand for electricity.

(4) The current household distributed PV consumption problem has a diversified subject of interest; distribution system operators, load aggregators, and individual users

have become the subject of interest, and how to maximize the benefits among multiple subjects is a problem.

(5) Privacy leakage and communication inefficiency during voltage regulation in low voltage distribution networks.

(6) In the process of voltage regulation, the degree of coordination of source, network, load, and storage flexibility resources is insufficient, which makes it difficult to realize the economic optimization and the best voltage regulation effect at the same time and the consideration of multiple flexibility resources for voltage regulation is accompanied by the complication of mathematical modeling and the increase in the difficulty of solving the problems.

4.2. Recommendations to Address the above Issues

(1) Consider the regulation sequence and economy of coordinated control of multiple voltage regulation methods. For the voltage problem caused by a high percentage of PV access, considering the combination of multiple regulation methods can achieve the purpose of cost reduction and efficiency improvement. Further research and additions should be made in terms of improving the degree of coordination of the source network and load, the regulation sequence of each device and the economics of regulation, and the control model that considers economic factors.

(2) Accurately obtain basic data and fully consider PV operation scenarios and distribution network operation indexes. Further research is necessary to establish a multi-scenario, multi-objective control model in low-voltage distribution networks containing a high proportion of household PV power generation, taking into account the suppression of network risk and optimization of network operation indexes, accurately obtaining basic data, reasonably constructing the model, and improving the accuracy of the analysis results.

(3) Focus on new power electronic equipment low-voltage distribution network structure transformation. Solid-state tap transformers and other new power electronic equipment have superior characteristics. The opening and closing operation of the thyristor can be adjusted to the transformer ratio to avoid the wear and tear of the taps and enhance the voltage regulation of the distribution network. The low-voltage AC distribution network transitioned to a low-voltage AC and DC hybrid distribution network in the current trend of network development. The low-voltage distribution network basically used a no-load voltage regulator transformer. For the distribution network containing distributed photovoltaic PV penetration rate will be further increased in the future, the transition of a low-voltage AC distribution network to a low-voltage AC-DC hybrid distribution network should be considered to carry out the OLTC transformation of low-voltage distribution network and the application of new type of power electronic devices, which can jointly complete the voltage regulation of low-voltage distribution network and ensure the quality of power supply to the users.

(4) Considering load-side demand response and enhancing the economic and price attributes of voltage control in low-voltage distribution networks. Currently, the state has released a number of policies that are favorable to distributed PV construction, but ultimately, distributed PV construction will be regulated by the market mechanism, and the subject of interest in household PV consumption will become diversified. It can be considered to establish a game and cooperation model of PV grid integration among different interests, and related research will play a role in promoting PV grid integration. For example, the distribution system operator influences the time of use of electricity by subsidizing the reduction of the frequency of voltage violations. Load aggregators help users schedule demand during the subsidy period, and individual users achieve low electricity prices to maximize the profit of each subject.

(5) Focus on privacy protection techniques, enhance communication efficiency, and consider artificial intelligence algorithms to cope with voltage overruns in distribution networks. With the increasing share of distributed PV in the distribution network, voltage overrun solutions combining front-end technologies and existing regulation methods need

to be considered. The literature [75] considers the impact of distributed renewable energy access on the voltage of low-voltage distribution networks and proposes a deep reinforcement learning-based scheduling strategy for energy storage devices. The combination of deep reinforcement learning algorithm and energy storage device scheduling is utilized to mitigate the voltage overrun problem. The literature [76] proposes a reactive power optimization method for distribution networks based on multi-intelligent body deep reinforcement learning. Multi-intelligent body DDPG is used to coordinate the control of multiple controllable devices such as capacitors, on-load voltage regulator transformers, and distributed power supplies. The literature [77] proposes a data-driven distributed control method based on coherent multi-intelligence deep reinforcement learning, where each inverter is modeled as an intelligent body, each intelligent body is responsible for the controllable devices in a region, and the intelligences communicate with each other to improve the coordination of control commands. It has high communication efficiency and robustness to communication faults. Further exploring the application of front-end technology in distribution network voltage management is one of the current research hotspots, such as artificial intelligence, deep reinforcement learning, etc., to improve the distribution network voltage regulation capability.

Voltage control methods applying multi-intelligent body learning, as mentioned in the literature [78], consider the latest learning algorithms for voltage control in order to drastically reduce the communication requirements and better protect the user's privacy to fulfill the current requirements of multi-source data fusion techniques considering privacy protection.

(6) For the problem of insufficient coordination of source, network, load, and storage flexibility resources, a comprehensive system model covering the characteristics, constraints, and interactions of different resources is needed for effective coordinated scheduling, which is still one of the research hotspots. Distributed control strategies can be used to realize the coordinated management of flexible resources. The system is decomposed into multiple subsystems, and distributed control algorithms are implemented on each subsystem to achieve optimal management of the overall system. Advanced optimization algorithms are used to achieve the goals of economic optimization and the best voltage regulation effect.

5. Conclusions

This review addresses the voltage overrun problem caused by high penetration distributed photovoltaic (PV) access to low-voltage distribution networks (LVDNs) and comprehends domestic and international research on voltage control methods for LVDNs.

Firstly, the working principles of four voltage control methods, namely, energy storage system configuration, regulating reactive power output of PV inverters, limiting active power output of PV, OLTC, and distribution network reconfiguration, are introduced, and the characteristics of each method are presented in the form of a table, which summarizes the advantages and defects of each method, and the four types of methods are compared and analyzed according to different criteria. The use of a single voltage regulation method has the problem of insufficient advantages and obvious defects, which is insufficient to meet the requirements of high penetration rate distributed PV grid connection in most cases, and the coordinated voltage control through multiple flexibility resources has become the current mainstream voltage regulation scheme.

After that, voltage regulation is considered from the perspective of flexibility resources, and the four voltage control methods in the previous section have been introduced for the three types of flexibility resources, i.e., considering distributed photovoltaic (PV) voltage regulation by itself, OLTC tap action and distribution network reconfiguration regulation, and energy storage system configuration regulation. The three types of flexibility resources, namely, source, network, and storage, have been widely used in distribution network voltage regulation, while load-side resources have become one of the main regulation resources of the new type of power system, but the current study introduces less about the participation of load-side flexibility resources in LV distribution network voltage regu-

lation, and advancing the application of load-side resources for voltage regulation in LV distribution networks is the focus of future research.

Then, the important role of load-side resources for voltage regulation is described in three parts, namely, the important role of load-side resources, the development trend of load-side resources, and the suggestions for promoting the coordination of source-network-load-storage flexibility resources, aiming to promote the application of load-side resources for voltage regulation in LV distribution networks and to combine them with the three types of flexibility resources of source-network-storage to achieve the purpose of controlling the cost and improving the efficiency of voltage regulation.

Finally, recommendations and solutions are presented to address the technical challenges of voltage regulation in LV distribution networks containing high penetration distributed PV. In the context of today's new power systems emphasizing the interaction of source, grid, load, and storage, new technologies and methods for solving the voltage problems in LV distribution networks are envisioned, with the hope that this article can provide some references for future research in this field.

Author Contributions: J.Z.: Conceptualization, Methodology, and Writing—review and editing. X.G.: Investigation, Extensive literature review, Writing—original draft. H.S.: Methodology, Writing—review and editing. Y.L.: Writing—review and editing. L.W.: Conceptualization and Methodology. C.Y.: Conceptualization and Methodology. Y.X.: Conceptualization and Methodology. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Scientific Research Program for Colleges and Universities in Hebei Province (QN2022028).

Conflicts of Interest: The funders had no role in the writing of the manuscript. The authors declare no conflicts of interest.

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